



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
29-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

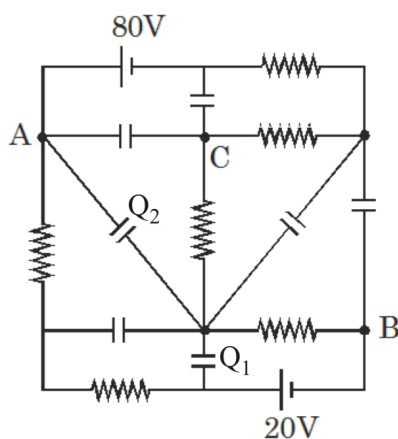
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

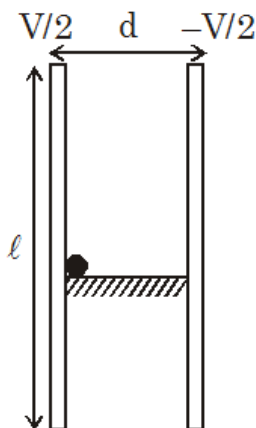
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For the given circuit, choose the **CORRECT** statements: (given all capacitors have capacitance = 1 F & all resistors have resistance = 10 Ω)

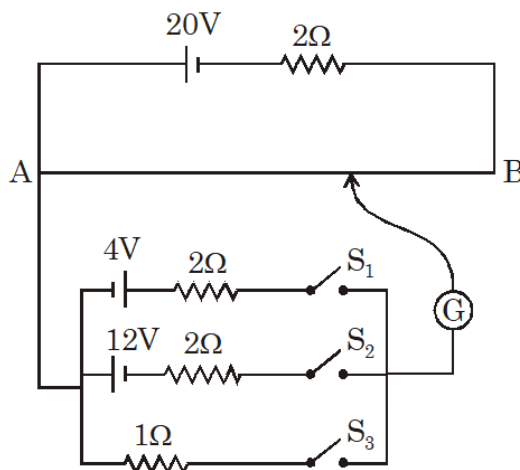


- (A) Potential difference between point A & B ($V_A - V_B$) is 40 V.
- (B) Potential difference between point A & B ($V_A - V_B$) is 50 V.
- (C) Magnitude of charge Q_1 is 30 C.
- (D) Magnitude of charge Q_2 is 50 C.

2. Two large square metallic plates ($\ell \times \ell$) are separated by a distance d . The potential of plates remain constant at $\frac{V}{2}$ & $-\frac{V}{2}$. The plates are connected by a smooth horizontal insulating platform. A very small metallic ball of mass m and radius r can move on the platform. The collisions are perfectly inelastic. ($\ell \gg d \gg r$)

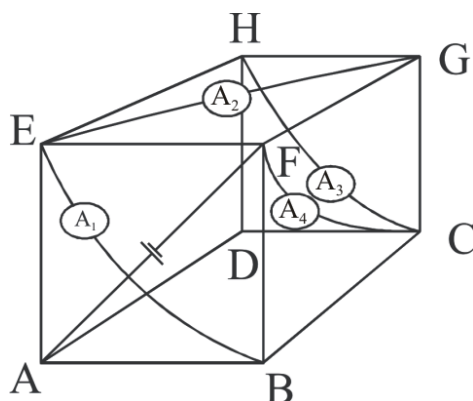


- (A) The effective charge that flows from left plate to right plate in a complete round trip is $4\pi\epsilon_0 V r$.
- (B) The average current over a large period of time is proportional to $V^{3/2}$.
- (C) The "resistance" to current flow $\left(\frac{V}{i}\right)$ is inversely proportional to V .
- (D) The average power loss is proportional to V^2 .
3. A potentiometer is connected to the circuit as shown in figure. Length of wire AB is 200cm and resistance is 8Ω .

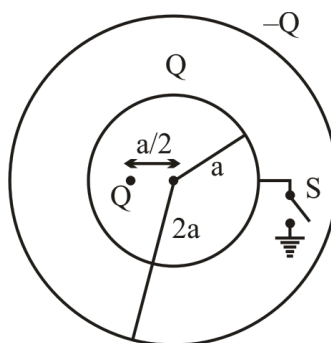


- (A) When only switch S_1 is closed null point is not obtained.
- (B) When only S_2 is closed null point is at 150 cm from A.
- (C) When both S_1 and S_2 are closed null point is not obtained.
- (D) When S_1 , S_2 and S_3 all are closed null point is at 50 cm from A.

4. ABCDEFGH is a cube made up of wire frame, there is equal Resistance R in each side of cube. Four identical ideal ameters are connected between vertices BE, EG, HC and FC as shown in figure. An ideal cell of EMF V is connected between A & F. Connections are only at vertices of cube. Choose correct readings of ameters.



- (A) Reading of A_1 is $3V/9R$ (B) Reading of A_2 is $2V/9R$
 (C) Reading of A_3 is $7V/9R$ (D) Reading of A_4 is $14V/9R$
5. Two concentric conducting thin shell of radii a and $2a$ having charge Q and $-Q$ respectively kept as shown in figure. Another point charge Q also kept at a distance $\frac{a}{2}$ from common center, then



- (A) Energy stored in electric field from $r = a$ to $r = 2a$ is $\frac{Q^2}{4\pi\epsilon_0 a}$, where r is radial distance from common center.
 (B) Electric potential of inner shell is $\frac{3Q}{8\pi\epsilon_0 a}$.
 (C) If switch S is closed, then charge flow through the switch into the earth is $\frac{5Q}{2}$.
 (D) Electric field at common center remains unchanged on closing the switch S .

6. A particle of mass m and charge $(-q)$ is constrained to move in the horizontal plane. The particle can oscillate back and forth in a straight line. A charge particle $(Q \gg q)$ is placed at a distance R below the origin of the horizontal plane and another charge $(-q)$ is fixed at the origin of the horizontal plane.

(A) The distance from origin at which the particle is in equilibrium is $r_0 = R \left(\frac{q}{Q} \right)^{\frac{1}{3}}$.

(B) The distance from origin at which the particle is in equilibrium is $r_0 = R \left(\frac{3q}{Q} \right)^{\frac{1}{3}}$.

- (C) The angular frequency of oscillations for small displacements around the equilibrium position is

$$\sqrt{\frac{qQ}{4\pi\epsilon_0 m R^3}}.$$

- (D) The angular frequency of oscillations for small displacements around the equilibrium position is

$$\sqrt{\frac{3qQ}{4\pi\epsilon_0 m R^3}}.$$

SECTION-I (ii) : (Maximum Marks: 12)

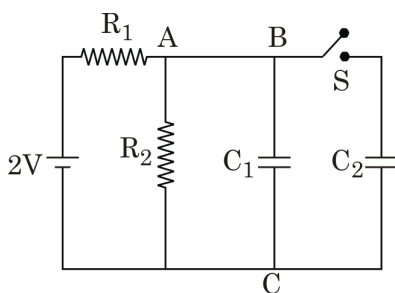
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

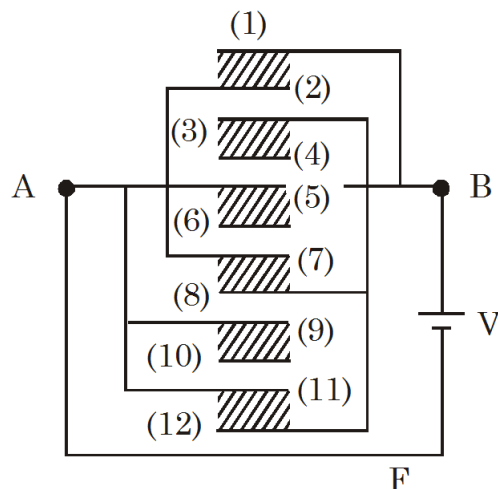
7. Initially the switch 'S' is open for a long time, Now the switch 'S' is closed at $t = 0$ ($R_1 = 2\Omega$, $R_2 = 2\Omega$, $C_1 = 1F$ and $C_2 = 1F$).



List-I		List-II	
(I)	The charge on capacitor C_1 (in C) just before the switch is closed	(P)	$\frac{1}{2e}$
(II)	The charge on capacitor C_1 (in C) just after the switch is closed	(Q)	$1 - \frac{1}{2e}$
(III)	The current through AB (in A) at $t = 2$ sec is	(R)	1
(IV)	The charge on capacitor C_1 (in C) long time after the switch is closed is	(S)	$\frac{1}{2}$
		(T)	None

- (A) I $\rightarrow$ R; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R
 (B) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ R
 (C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ P
 (D) I $\rightarrow$ R; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S

8. All identical capacitor plates each of area A are arranged such that adjacent plates are at d distance apart. Plates are connected to a source of emf V volts as shown in figure. Match the quantities in List -I with their respective answer in List-II. (dielectric constant in each shaded region is $k = 2$)



	List-I		List-II
(I)	Charge on plate 3 (multiple of $\frac{\epsilon_0 AV}{d}$)	(P)	4/3
(II)	Energy stored in the system (multiple of $\frac{13\epsilon_0 AV^2}{d}$)	(Q)	1/3
(III)	Equivalent capacitance between AB (multiple of $\frac{13A\epsilon_0}{d}$)	(R)	2/3
(IV)	Potential difference between the plates 4 and 7 (multiple of $V/2$)	(S)	5/3
		(T)	None

- (A) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ Q
 (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R
 (C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q
 (D) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ P

9. Answer the following by appropriately matching the lists based on the information given in the paragraph

A particle is projected vertically up from the surface of the earth with an speed $\sqrt{gR}$. Which of the following is the only correct combination ?

List-I		List-II	
(I)	Maximum height attained by the particle is αR the value of α is	(P)	1
(II)	The duration for which the particle is between the point of projection and half of maximum height (including upward and downward journey) is $\beta \sqrt{\frac{R^3}{GM}}$, the value of β is	(Q)	2
(III)	Time taken by the particle to travel from half of maximum height to maximum height is $\gamma \sqrt{\frac{R^3}{GM}}$, the value of γ is	(R)	$\frac{\pi}{3} + \frac{\sqrt{3}}{2}$
(IV)	The acceleration due to gravity at maximum height is $\frac{\delta GM}{4R^2}$, the value of δ is	(S)	$\left(\frac{\pi}{3} - \sqrt{3} + 2\right)$
		(T)	None

- (A) I $\rightarrow$ P; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ P
 (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R
 (C) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ P
 (D) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P

10. Electric field in a region is given by $\vec{E} = \left[(2xy)\hat{i} + (x^2)\hat{j} \right]$ V/m. Electric potential at origin is 5V. Match the following two columns.

List-I		List-II	
(I)	Potential at point (5m, 2m, 1m)	(P)	4V
(II)	Potential at point (1m, 1m, 1m)	(Q)	1V
(III)	Potential at point (2m, 1m, 1m)	(R)	3V
(IV)	Potential at point (1m, 2m, 1m)	(S)	-45V
		(T)	None

- (A) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ Q
 (B) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ Q
 (C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q
 (D) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R

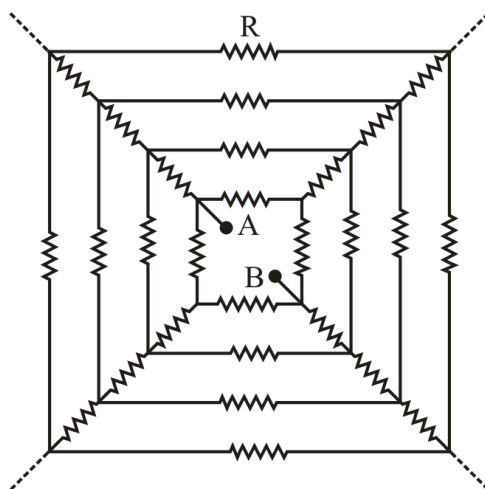
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

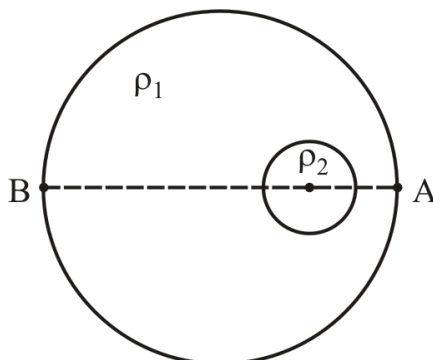
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

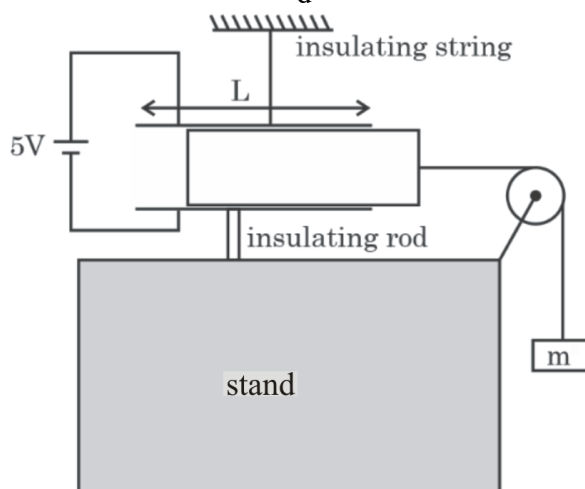
1. The circuit shows the pattern of "infinite" connection of resistances. Each resistance has a value $R = 1\Omega$. The equivalent resistance between A and B is



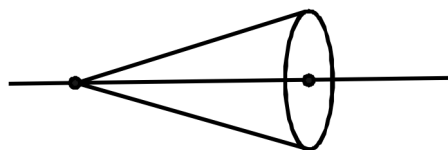
2. A spherical cavity of radius $\frac{R}{4}$ is cut from a sphere of radius R of density ρ_1 . The center of cavity is at a distance $\frac{R}{2}$ from the center of the sphere. The cavity is filled with a material of density ρ_2 , as shown in diagram. The acceleration due to gravity at point A is 10 % more than acceleration due to gravity at point B, the value of $\frac{\rho_1}{\rho_2}$ is



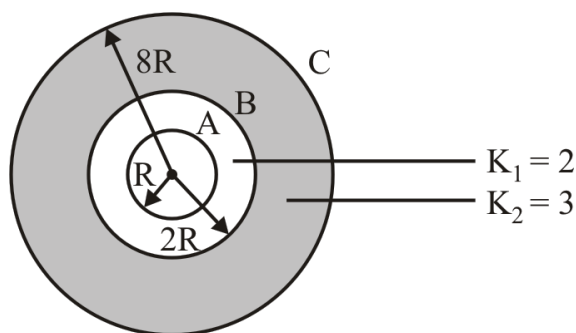
3. Separation between the plates of capacitor is d and cross section area of plates is Lb . Dielectric constant is 4. The dielectric is connected to a block of mass m as shown in the diagram. The value of m (in kg) for which the block moves with a constant velocity is. Given : $\frac{\epsilon_0 b}{d} = 10$ (in SI unit)



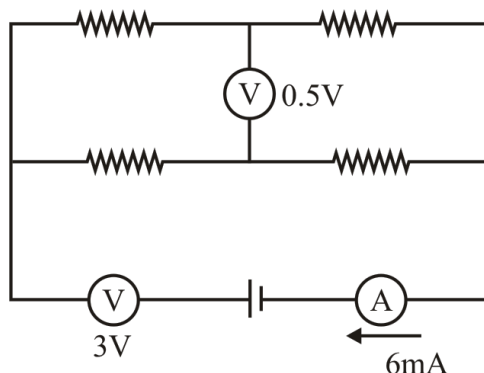
4. A cone with base radius $\sqrt{3}R$ & height R has charge q and $2q$ placed at vertices & base centre respectively. The total electric flux passing through cone is $x \frac{q}{\epsilon_0}$. Find x .



5. A long cylindrical capacitor has the cross section shown in figure. The cylinder of radius R has charge density σ . Rest are uncharged. Find V_{AB}/V_{BC} .



6. The range of ammeter can be changed by connecting a resistance box ABCD across the galvanometer. The terminals AD are connected across the galvanometer terminals. When the terminals AC are shorted, the range of ammeter is 4A. When terminals BD are shorted, the range of ammeter becomes 3A. When terminals BC are shorted, the range of ammeter becomes 2A. The full deflection current of the galvanometer is
7. Two small dipoles having dipole moments $p\hat{i}$ and $\sqrt{3}p\hat{j}$ are placed at the origin. The point on the circle of radius R, where the potential is maximum is (x, y). The ratio $\frac{y}{x}$ is
8. Four resistance R, 3R, R and 3R are connected as shown in figure. The two voltmeters are identical and Ammeter is ideal. The readings of ammeter and voltmeter are shown in the diagram. The value of R is.



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Which of the following statement is/are **CORRECT**?
 - (A) Higher is the gold number of a lyophilic sol more is its coagulating power
 - (B) An electrolyte can coagulate both positive sol and negative sol
 - (C) All impurities present in a colloid makes it more stable
 - (D) Dialysis is a process to remove impurities of ions and smaller molecules from a colloid
 2. Kolbe's electrolysis of sodium acetate was carried out in aqueous medium. Given 75% yield of electrolysis process 4.4 gm CO_2 is produced at anode. Select correct option(s) :
 - (A) Total charge passed in $8/3F$
 - (B) Mass of hydrocarbon produced is 1.5 gm
 - (C) Total mass of gases produced during the process is 6.1 gm
 - (D) Total mass of gases produced during the process is 6 gm

3. At 298 K, the standard reduction potentials are 1.51 V for $\text{MnO}_4^-|\text{Mn}^{+2}$, 1.36 V for $\text{Cl}_2|\text{Cl}^-$, 1.07 V for $\text{Br}_2|\text{Br}^-$, and 0.54 V for $\text{I}_2|\text{I}^-$. At pH = 3, permanganate is expected to oxidize $\left(\frac{RT}{F} = 0.059 \text{ V}\right)$: –
- (A) Cl^- (B) Br^-
(C) I^- (D) All Cl^- , Br^- , I^-
4. Which is/are correct statement(s)
- (A) A catalyst may appear in the kinetic rate law equation of the reaction
(B) A reaction having first order may be either elementary or complex.
(C) A plot of rate of reaction v/s concentration of reactant for first order reaction ($\text{A} \rightarrow \text{Product}$) is a straight line with positive slope
(D) A zero order reaction may be elementary or complex.
5. Which of the following statement(s) is(are) correct ?
- (A) Smoke and cloud both are aerosols with same dispersion medium.
(B) Surface tension of lyophilic sols is less than the surface tension of dispersion medium while that of lyophobic sols is more than that of dispersion medium.
(C) Lyophobic sols can be observed using ultra microscope
(D) Colloidal sol shows low magnitude of colligative properties, as compared to true solutions.
6. For the reaction at 273°C
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
 It is observed that the pressure of $\text{H}_2(\text{g})$ falls from 760 mmHg to 380 mmHg in 100 sec.
 Select the correct statement(s).
- (A) Average rate of reaction is $5 \times 10^{-3} \text{ atm/sec}$.
(B) Average rate of formation of $\text{NH}_3(\text{g})$ is $\frac{1}{300} \text{ atm/sec}$.
(C) Average rate of disappearance of $\text{N}_2(\text{g})$ is twice of rate of appearance of $\text{NH}_3(\text{g})$.
(D) $\log\left(\frac{-\Delta P_{\text{H}_2}}{\Delta t}\right) = \log\left(\frac{-\Delta P_{\text{N}_2}}{\Delta t}\right) + 0.4771$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I Electrode (25° C 1atm)		List-II Oxidation potential (approximate)	
(P)	Pt, H ₂ NaCl(aq)	(1)	0.66 volts
(Q)	Pt, H ₂ NH ₄ Cl(K _b of NH ₃ = 10 ⁻⁶) (1M)	(2)	0.6 volts
(R)	Pt, H ₂ HCOOK(K _b of HCOO ⁻ = 10 ⁻⁶) (1M)	(3)	0.24 volts
(S)	Pt, H ₂ KSH (K _{a1} =10 ⁻⁴ , K _{a2} =10 ⁻¹⁰)	(4)	0.42 volts
		(5)	0.36 volt

[Given : $\frac{2.303 RT}{F} = 0.06$]

The correct match is :

- (A) P → 4; Q → 4; R → 1; S → 5 (B) P → 4; Q → 2; R → 3; S → 4
- (C) P → 4; Q → 3; R → 2; S → 2 (D) P → 4; Q → 3; R → 1; S → 4

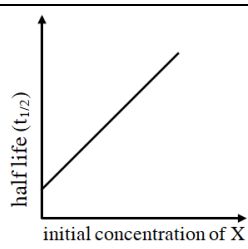
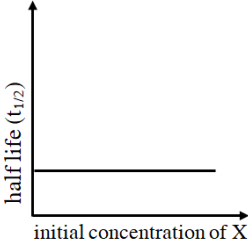
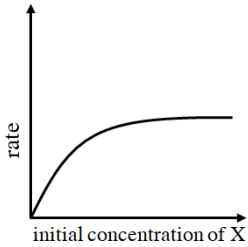
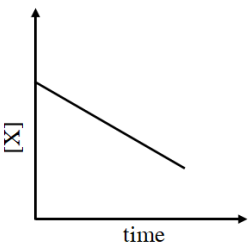
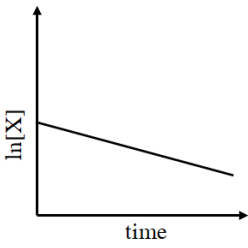
8. Match the List-I with List-II.

List-I		List-II	
(P)	Molten PbCl ₂ using inert electrode	(1)	Metal of salt will be reduced
(Q)	NaCl aqueous solution using inert electrode	(2)	2H ₂ O + 2e ⁻ → H ₂ (g) + 2OH ⁻
(R)	Silver nitrate aqueous solution using Ag electrodes	(3)	Solution becomes basic after electrolysis
(S)	Sodium nitrate aqueous solution using inert electrodes	(4)	Solution becomes acidic after electrolysis
		(5)	Solution becomes neutral after electrolysis

[Ignore hydrolysis of ions in solution]

- (A) P → 1; Q → 3; R → 5; S → 2 (B) P → 1; Q → 2; R → 1; S → 3
- (C) P → 5; Q → 5; R → 4; S → 5 (D) P → 5; Q → 2; R → 4; S → 5

9. Match the rate expressions in LIST-I for the decomposition of X with the corresponding profiles provided in LIST-II. X_s and k constants having appropriate units.

LIST-I		LIST-II	
(P)	$\text{rate} = \frac{k[X]}{X_s + [X]}$ under all possible initial concentration of X	(1)	
(Q)	$\text{rate} = \frac{k[X]}{X_s + [X]}$ where initial concentration of X are much less than X_s	(2)	
(R)	$\text{rate} = \frac{k[X]}{X_s + [X]}$ where initial concentration of X are much higher than X_s	(3)	
(S)	$\text{rate} = \frac{k[X]^2}{X_s + [X]}$ where initial concentration of X is much higher than X_s	(4)	
		(5)	

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 5$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 5$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

10. Which of the following option show correct matches :-

List-I		List-II	
(P)	Precipitation of colloidal sol	(1)	Peptization
(Q)	Purification of colloid	(2)	Coagulation
(R)	Gold Number	(3)	Desorption
(S)	Formation of colloidal sol	(4)	Dialysis
		(5)	Measure of protective power

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(B) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Consider the following statements:
 - Osmotic pressure of rubber sol will be same as that of sucrose solution having same mass of rubber and sucrose mixed in same mass of H_2O
 - Soap sol of sodium palmitate will be coagulated near cathode on electrophoresis.
 - For miscellisation of surfactants entropy of system increases.
 - Movement of dispersion medium under the influence of an electric field is called cataphoresis.
 - Gold sol is a negatively charged hydrophilic sol.
 - Tyndall effect is more distinct for lyophobic colloid than for lyophilic colloids.
 - Gold sol on mixing with starch sol causes stabilisation of starch sol.
 - During adsorption there will be an increase in residual forces on the molecules.
 - The extent of adsorption per unit mass of adsorbent will remain same even on changing surface area of same mass of adsorbent.
 - On adding 0.1 M KI solution slowly to 0.01 M $AgNO_3$ solution, causes conductivity to increase gradually and after some time it becomes constant.

If number of correct statements = x
 And number of incorrect statements = y
 Then, $\frac{x}{y} =$
- If "X" represents reduction potential (E^0 , in V) of $MnO_4^- (aq)/Mn(s)$ and "Y" represents reduction potential (E^0 , in V) of $Cr_2O_7^{2-} (aq)/Cr(s)$ then $|X-Y|$ is.
 [Given : $E^0_{(MnO_4^- (aq)/MnO_2(s))} = 1.7V$; $E^0_{(MnO_2(s)/Mn^{2+}(aq))} = 1.25V$; $E^0_{(Mn^{2+}(aq)/Mn(s))} = -1V$]
 $[E^0_{(Cr_2O_7^{2-} (aq)/Cr^{3+}(aq))} = 1.36V$; $E^0_{(Cr^{3+}(aq)/Cr(s))} = -0.74V$]
- The conductivity of 0.001M Na_2SO_4 solution is $2.6 \times 10^{-4} S cm^{-1}$ and it increases to $8.2 \times 10^{-4} S cm^{-1}$, when the solution is saturated with $CaSO_4$. The limiting molar conductances of Na^+ and Ca^{2+} ions are 50 and $120 S cm^2 mol^{-1}$, respectively. Neglecting conductivity of water, the solubility product of $CaSO_4$ is $Y \times 10^{-x} M^2$. Then Y is.

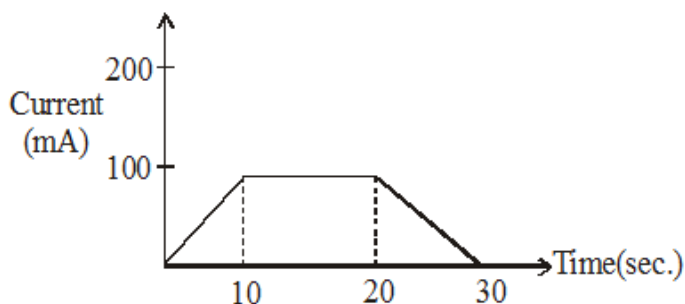
4. The decomposition of A_2B_2 follows the rate equation $r = K[A_2B_2]^2$.

$A_2B_2 \xrightarrow{K} A_2 + B_2$. The kinetic data of the reaction is given below.

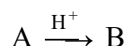
Initial conc. of A_2B_2 (mole/lit)	7
First half life of A_2B_2 in sec.	1440

If the initial concentration of A_2B_2 were 56 m mol/lit. and the time taken for the conversion of A_2B_2 to reach 60% of initial value is 't' seconds then find value of $\left(\frac{t}{1000}\right)$?

5. If mass of copper deposited in a copper voltameter is 30 grams in 30 seconds. The time current graph is as shown in figure, calculate electrochemical equivalent (gm / coulomb) of Copper during the entire process.



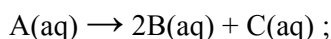
6. For any acid catalysed reaction



half life period is independent of concentration of A at given pH. At given concentration of A, half life period is 10 min at pH = 2 and half life time is 1000 min at pH = 3. If the rate law expression of reaction is :

Rate = $K[A]^x [H^+]^y$, then calculate value of $(x + y)$?

7. For the first order reaction



where A is an optically active compound while B is reducing agent which reacts with H_2O_2 . The progress of reaction is monitored by measuring optical rotation as well by titrating a definite volume of reaction mixture with H_2O_2 from time to time. In an experiment, optical rotation was found to be $\theta = 30^\circ$ at $t = 20$ min and $\theta = 15^\circ$ at time $t = 50$ min from the start of reaction. If in the same experiment volume of H_2O_2 consumed at $t = 30$ min from start is 30 mL then volume of H_2O_2 which is consumed by same volume of reaction mixture at $t = 90$ min will be :

[Compound 'B' and 'C' are optically inactive]

8. 10% sites of catalyst bed has been occupied by H_2 gas molecules. On heating H_2 gas is evolved from sites and collected at 0.030 atm and 300K in a vessel of volume 2.46 cm^3 . Total number of sites available on catalyst is 6×10^{15} per cm^2 and surface area of catalyst is 10^3 cm^2 . The number of H_2 molecules adsorbed per site of catalyst are :

(Given $N_A = 6 \times 10^{23}$, $R = 0.082 \text{ L.atm K}^{-1} \text{ mol}^{-1}$)

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $f : [0, 1] \rightarrow \mathbb{R}$ be a function such that $(f(x))^2 = 1$ and $f(x)$ is discontinuous at exactly two points and

$$\int_0^1 f(x)dx = 0. \text{ Let } I = \int_0^1 xf(x)dx, \text{ choose the **incorrect** option(s)}$$

(A) maximum value of I is $\frac{1}{4}$

(B) minimum value of I is $-\frac{3}{4}$

(C) maximum value of I is 3

(D) minimum value of I is -2

2. Let $S_n = \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \frac{1024}{n^6} + \dots + \frac{1}{n} \right)$ and $T_n = \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \frac{1024}{n^6} + \dots + \frac{(n-1)^5}{n^6} \right)$, then which of the following is/are true ?

(A) $S_n > \frac{1}{6}$

(B) $(S_n + T_n) < \frac{1}{3}$

(C) $(S_n + T_n) > \frac{1}{3}$

(D) $T_n < \frac{1}{6}$

3. The solution of $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying $y(1) = 1$ is given by :
- (A) a hyperbola
(B) a circle
(C) $y^2 = x(1 + x) - 1$
(D) $(x - 2)^2 + (y - 3)^2 = 5$
4. If $\int_2^3 \frac{e^{x-1}}{(x^2 - 5x + 4)} 2x dx = AF(1) + BF(-2)$ and $F(t) = \int_t^{t+1} \frac{e^x}{x} dx$, then
- (A) $A = -2/3$
(B) $B = (4/3)e^3$
(C) $A = 2/3$
(D) $B = (8/3)e^3$
5. Let R be the region in the first quadrant bounded by the x-axis, the line $2y = x$, and the ellipse $\frac{x^2}{9} + y^2 = 1$. Let R' be the region in the first quadrant bounded by the y-axis, the line $y = mx$, and the given ellipse, then the value of m such that R and R' have the same area is
- (A) $\frac{2}{9}$ (B) $\frac{3}{11}$
(C) $\frac{2}{5}$ (D) $\frac{2}{7}$
6. If a function $y = f(x)$ is such that $f'(x)$ is continuous function and satisfies $(f(x))^2 = K + \int_0^x ((f(t))^2 + (f'(t))^2) dt$, $K \in \mathbb{R}^+$, then which of the following is/are possible ?
- (A) $f(x)$ is increasing function $\forall x \in \mathbb{R}$
(B) $f(x)$ is bounded function
(C) $f(x)$ is neither even nor odd
(D) if $K = 100$ then $f(0) = 10$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. A function F is defined by $F(x) = \int_1^x \frac{e^t}{t} dt \forall x > 0$. Now express the functions in List-I in terms of F , then match the following List-I with List-II

List-I		List-II	
(I)	$\int_1^x \frac{e^t}{t+2} dt$	(P)	$F(x) - \frac{e^x}{x} + e$
(II)	$\int_1^x \frac{e^{3t}}{t} dt$	(Q)	$xe^{1/x} - e - F\left(\frac{1}{x}\right)$
(III)	$\int_1^x \frac{e^t}{t^2} dt$	(R)	$e^{-2}[F(x+2) - F(3)]$
(IV)	$\int_1^x e^{1/t} dt$	(S)	$F(3x) - F(3)$
		(T)	$F(3x) + F(3)$

- (A) $I \rightarrow S; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$
 (B) $I \rightarrow R; II \rightarrow S; III \rightarrow P; IV \rightarrow Q$
 (C) $I \rightarrow R; II \rightarrow S; III \rightarrow Q; IV \rightarrow P$
 (D) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$

8. Match the following List-I with List-II (C is integration constant)

List-I		List-II	
(I)	$\int \frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} dx$	(P)	$\ln \left(\frac{(x^2 + 1) + \sqrt{x^4 + 1}}{x} \right) + C$
(II)	$\int \frac{x^2 - 1}{x \sqrt{1 + x^4}} dx$	(Q)	$C + \frac{1}{\sqrt{2}} \ln \left(\frac{\sqrt{x^4 + 1} - \sqrt{2}x}{(x^2 - 1)} \right)$
(III)	$\int \frac{1 + x^2}{(x^2 - 1)\sqrt{1 + x^4}} dx$	(R)	$C - \tan^{-1} \left(\sqrt{\sqrt{1 + \frac{1}{x^4}} - 1} \right)$
(IV)	$\int \frac{dx}{(1 + x^4)\sqrt{\sqrt{1 + x^4} - x^2}}$	(S)	$\frac{\sqrt{x^4 + x^2 + 1}}{x} + C$
		(T)	$\ln \left(\frac{(x^2 + 1) - \sqrt{x^4 + 1}}{x} \right) + C$

(A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S

(B) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R

(C) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ S

(D) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ Q

9. Match the following List-I with List-II

List-I		List-II	
(I)	$xdy = y(dx + ydy)$, $y(1) = 1$ and $y(x_0) = -3$, then $x_0 = \lambda$, then λ is less than	(P)	$\frac{1}{4}$
(II)	If $y(t)$ is solution of $(t + 1)\frac{dy}{dt} - ty = 1$, $y(0) = -1$, and $y(1) = \lambda$, then λ is greater than	(Q)	-15
(III)	$(x^2 + y^2)dy = xydx$, $y(1) = 1$, $y(x_0) = \sqrt{e}$ and $x_0 = \lambda$, then λ is greater than	(R)	$-\frac{1}{2}$
(IV)	$\frac{dy}{dx} + \frac{2y}{x} = 0$, $y(1) = 1$, then $y(2)$ is equal to	(S)	$\sqrt{3}e$
		(T)	1

(A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ Q

(B) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P

(D) I $\rightarrow$ P; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ Q

10. Match List-I with List-II :

List-I		List-II	
(I)	The area bounded by $2^{ x } y + 2^{ x -1} \leq 1$ is	(P)	$4 - \frac{2}{\ln 2}$
(II)	The area bounded by $2^{ x } y + 2^{ x -1} \leq 1$, $ x \leq \frac{1}{2}$ & $ y \leq \frac{1}{2}$ is equal to	(Q)	$\frac{2}{\ln 2} - 2$
(III)	The area bounded by $2^{ x } y + 2^{ x -1} \geq 1$ & $ x + y \leq 1$ is equal to	(R)	$\frac{4(2^{1/3} - 1)}{2^{1/3} \ln 2} - \frac{2}{3}$
(IV)	The area bounded by $2^{ x } y + 2^{ x -1} \leq 1$ & $ x \leq \frac{1}{3}$	(S)	$\frac{(\sqrt{2} - 1) 2\sqrt{2}}{\ln 2} - 1$
		(T)	$\frac{2}{\ln 2} + 2$

(A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S

(B) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ R

(D) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ Q

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $f(x + f(y)) + f(x - f(y)) = f(x) \forall x, y \in \mathbb{R}$ and $(0, 2)$ lies on $y = f(x)$, then $\int_0^{12} f(x) dx = \lambda \int_{-24}^{24} f(x - 36) dx$ where λ equals
2. If $\sum_{r=0}^{\infty} \left(\frac{1}{4r+1} - \frac{1}{4r+3} \right) = L$, then $[4L]$ is (where $[.]$ denotes greatest integer function)
3. If f and g are differentiable functions such that $xg(f(x))f'(g(x))g'(x) = f(g(x))g'(f(x))f'(x)$ for all real x . Also f is non-negative and g is positive. If $\int_0^p f(g(x)) dx = \frac{1}{2} - \frac{e^{-2p}}{2}$ for all real p and $g(f(0)) = 1$. If the value of $g(f(4))$ is e^{-k} , where $k \in \mathbb{N}$. Then k is
4. If the line $y = mx$ bisects the area enclosed by the lines $x = 0, y = 0, x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$ for $m = \frac{p}{q}$ ($p, q \in \mathbb{N}$, p & q are co-prime}, then the value of $2p - 3q =$

5. If $\int \frac{(x+2) dx}{\sqrt{x^4+8x^3+20x^2+16x+4}} = \frac{1}{2} \ln f(x) + c$, then $f(1)$ is equal to
6. Area of points satisfying $|2x+3y| + |3x-4y| \leq 2$ equals
7. Given ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and circle $C : (x-c)^2 + y^2 = c^2$. If E is of smallest possible area (Δ) such that it contains C , and $c = \frac{\sqrt{2}}{\sqrt{\sqrt{3}\pi}}$, then find Δ .
8. Let f be a continuous function that $p(x) = \int_{-2}^2 f(t)|x-t|dt$ where $x \in (-2, 2)$, then the value of $\int_0^2 \frac{p''(x)}{f(x)} dx$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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